Operation and Maintenance Manual

(Translation of the original Operation and Maintenance Manual)

KD27V12

From serial number 2016160001

KOHLER®

EN_US
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09-2022

KOHLER.

EMISSION CONTROL WARRANTY STATEMENT (USA)

For the United States only

EMISSION WARRANTY

Kohler warrants to the initial and subsequent owners of a stationary equipment certified diesel engine that such engine is:

- Designed, manufactured and equipped to conform at the time of sale with all applicable emission regulations adopted by the United States Environmental Protection Agency ("EPA") under Section 213 of the Clean Air Act.
- Free from defects in materials and workmanship with respect to the warranted emission control system and components for a period of five (5) years or 3000 hours of total operation by all persons, whichever event occurs first, after date of delivery to the initial owner.

If any warranted part that is scheduled for replacement as required maintenance fails prior to the first scheduled replacement point for that part, that part will be repaired or replaced by Kohler. If a component of the emission control system fails during the warranty period, it will be repaired or replaced under warranty and warranted for the remainder of the warranty period. Kohler will provide the repair or replacement of any component warranted to the stationary equipment engine owner. The warranty period begins on the date the engine (in service) or machine (in production) is delivered to the initial owner, and upon any sale of the engine by such initial owner to a subsequent owner, the warranty period does not restart, but rather the remaining balance of the warranty period transfers to the subsequent owner.

WARRANTED PARTS

The following parts (as equipped) are warranted under this warranty for all Kohler engines:

- Fuel Injection System
- Air Induction System
- Turbocharger System
- Charge Air Cooling System
- Exhaust Gas Recirculation (EGR) System
- EGR Control System
- Exhaust manifold
- Ignition System
- Diesel Particulate Filter System
- Diesel Oxidation Catalyst
- Fuel Additive Devices or Exhaust Aftertreatment Devices
- Selective Catalyst Reduction
- Reductant Containers
- Electronic Control Unit, Sensors, Solenoids, and Wiring Harnesses
- Emission Control Information Label
- Crankcase Ventilation Valves

RESPONSIBILITIES & LIMITATIONS

This warranty is subject to the following conditions.

Kohler's Responsibilities:

In case an emission-related defect is found in a warranted component during the warranty period, Kohler will provide:

- New, remanufactured, or properly repaired components, approved pursuant to applicable regulations, required to correct the defect.
- Reasonable and customary labor as established by Kohler, during normal working hours, required to make the warranted repairs, including labor to remove and install the engine, if so required.
- Parts replaced under this warranty become the property of Kohler.

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Owner's Responsibilities:

During the warranty period, the owner of the stationary equipment engine is responsible for:
- Premium or overtime labor costs.
- Costs to investigate engine problems which are not caused by a defect in Kohler's material or workmanship.
- Kohler may deny warranty coverage if the stationary engine or a part thereof has failed due to abuse, neglect, improper maintenance, or unapproved modifications.
- Providing timely notice of a warranted failure to an authorized Kohler Dealer, and to promptly make the equipment or engine available to the Kohler Dealer for repair.
- Proper scheduled and preventive maintenance as outlined in the Operation & Maintenance Manual supplied with the equipment. This includes, but is not limited to, valve adjustment, fuel and oil filter changes, and any other maintenance procedure related to emission control.
- Using the proper fuel in the engine, as specified in the Operator's Manual. Engines other than Tier 4 diesel engines shall only be operated on commercially available diesel fuel. Use of any other fuel may result in the engine no longer operating in compliance with EPA requirements. The Tier 4 diesel engine shall only be operated on ultra-low sulfur diesel fuel.
- The owner is responsible for initiating the warranty process. Owners are advised to contact their local authorized Kohler Dealer to perform warranty service as soon as a problem arises. The warranty repairs should be completed by the authorized Kohler Dealer as expeditiously as possible.

EXCLUSIONS

This warranty does not cover:
- **This warranty shall only apply** to KOHLER Co. engines.
- Malfunctions in any part caused by abuse, misuse, alterations, tampering, disconnection, or improper or inadequate maintenance.
- Attachments, accessories or components not supplied or approved by Kohler.
- Damage resulting from fire, accident, negligence, act of God or other events beyond the control of Kohler.
- Consequential damage such as loss of use of the engine or equipment powered by the engine, towing, machine transportation, loss of time, downtime, inconvenience, telephone, travel, lodging, or any other indirect or direct damage.
- Loss or damage to personal property, loss of revenue, commercial loss or any other matters not specifically included in this warranty statement.
- Any warranted part that was required to be previously replaced as part of required scheduled maintenance.
- Owner's unreasonable delay in making the equipment available after being notified of a potential product problem.
- Engines installed outside United States of America.

This warranty is in addition to the Kohler Standard Warranty and any extended warranty (if applicable) for the equipment and engine involved. Remedies under this warranty are limited to the provisions as specified in this warranty statement.

IN NO EVENT SHALL KOHLER, ITS COMPONENT SUPPLIERS NOR THE SELLING DEALER BE LIABLE FOR ANY DELAY, WORK STOPPAGE, LOSS OF USE OF EQUIPMENT, LOSS OF TIME, INCONVENIENCE, LOSS OF PROFITS, OR ANY INDIRECT, INCIDENTAL, SPECIAL, OR CONSEQUENTIAL DAMAGES RESULTING FROM OR ATTRIBUTABLE TO, DEFECTS IN KOHLER PRODUCTS OR SERVICE, WHETHER RESULTING FROM NEGLIGENCE (INCLUDING GROSS NEGLIGENCE) OR OTHER TORT WARRANTY, CONTRACT, INDEMNITY, BREACH OF THE PROMISE TO REPAIR OR REPLACE CONTAINED HEREIN, STRICT LIABILITY OR OTHERWISE.

IN ADDITION, KOHLER, ITS COMPONENT SUPPLIERS AND THE SELLING DEALER SHALL NOT BE LIABLE IN TORT OR STRICT LIABILITY FOR ANY ECONOMIC LOSS RESULTING, IN WHOLE OR IN PART, FROM THE MANUFACTURE OR SUPPLY OF THE PRODUCT, PARTS, COMPONENTS AND/OR LABOR.

NOTWITHSTANDING ANYTHING IN THIS WARRANTY TO THE CONTRARY, THE MAXIMUM LIABILITY, IF ANY, OF KOHLER, ITS COMPONENT SUPPLIERS AND THE SELLING DEALER FOR

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ALL DIRECT DAMAGES SHALL NOT EXCEED THE PRICE OF THE PARTICULAR DEFECTIVE PRODUCT, PART OR SERVICE, AS APPLICABLE.

THE STANDARD WARRANTY, THE EMISSION CONTROL WARRANTIES, AND THE EXTENDED LIMITED MACHINE AND POWERTRAIN WARRANTIES, IF APPLICABLE, ARE THE ONLY WARRANTIES APPLICABLE TO KOHLER PRODUCTS AND COMPONENTS USED IN KOHLER PRODUCTS AND ARE EXPRESSLY IN LIEU OF ANY WARRANTIES OR CONDITIONS OTHERWISE IMPLIED BY LAW (INCLUDING ANY WARRANTY OTHERWISE IMPLIED BY LAW FOR THE PRODUCT BY THE MANUFACTURER OR ITS COMPONENT SUPPLIERS), INCLUDING, BUT NOT LIMITED TO, IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE AND IS NOT A WARRANTY OF FUTURE PERFORMANCE.

THE REMEDIES UNDER THIS WARRANTY SHALL BE THE ONLY REMEDIES AVAILABLE TO THE OWNER OF KOHLER PRODUCTS OR ANY OTHER PERSON, AND NEITHER KOHLER NOR THE SELLING DEALER ASSUMES ANY OTHER OBLIGATION OR RESPONSIBILITY WITH RESPECT TO THE CONDITION OF KOHLER PRODUCTS, AND AUTHORIZES NO OTHER PERSON TO ASSUME FOR ANY OF THEM, ANY OTHER OBLIGATION OR LIABILITY.

Kohler reserves the right to make changes in design or add any improvements on its products at any time without incurring any obligation to install same on units previously delivered.

If the original owner sells the engine and machine, the remaining period of this warranty shall be transferred to the new owner.

“Kohler” means both Kohler Co. and its Affiliates, d/b/a Kohler Power Systems.

Kohler request that each owner of the engine and machine complete and provide the following requested information, sign in the space indicated, retain a completed and signed copy of this statement for the owner’s records, and return a completed and signed copy of this statement to Kohler. Failure to complete, sign, or return this form will not affect any owner’s rights or Kohler’s obligations under this warranty statement and will not result in denial of warranty coverage by Kohler.

MODEL: _____	PIN (Prod. Id. No.): _____
NAME OF DEALER	NAME OF OWNER
SIGNATURE OF AUTHORIZED REPRESENTATIVE OF DEALER	SIGNATURE OF AUTHORIZED REPRESENTATIVE OF OWNER
CITY / STATE / ZIP CODE OF DEALER	CITY / STATE / ZIP CODE OF OWNER
DATE MACHINE PLACED INTO SERVICE	

This Emission Control Warranty Statement is applicable to KD18L06, KD27V12, KD36V16, KD45V20, KD62V12, KD83V16 and KD103V20 Engines

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Preface

Only for the United States of America:

⚠ **WARNING:** This product can expose you to chemical substances, including carbon monoxide and benzene, which are known to the State of California to cause cancer, birth defects and reproductive harm.
Additional information at https://www.p65warnings.ca.gov/.

⚠ **WARNING:** Inhaling diesel engine exhaust exposes you to chemical substances, which are known to the State of California to cause cancer, birth defects and reproductive harm.
- Engines may only be started and operated in well ventilated areas.
- In closed rooms, the exhaust must be discharged into the open air.
- The exhaust system may not be changed or manipulated.
- Only let the engine run at an idle when it is necessary.

Additional information at https://www.p65warnings.ca.gov/.

This operation and maintenance manual has been written for the **operator** and **maintenance personnel** of the engine.

It contains descriptions for:
- Technical data
- Safety regulations
- Handling and operation
- Maintenance

The operation and maintenance manual is to be carefully read and applied before the first commissioning and later at regular time intervals by each person who is assigned to work with / on the engine.

Work with or on the engine is, for example:
- **Operation**
- **Servicing**, including maintenance, inspection

This facilitates familiarization with the engine for the operator and prevents malfunctions due to improper handling.

Kohler Co. will not accept any warranty claims that arise due to improper handling, insufficient maintenance, use of unapproved fuel and operating fluids, or failure to observe the safety regulations.

Kohler Co. will cancel all possible obligations undertaken by Kohler Co. and/or its dealers, such as guarantee commitments, service agreements etc., without prior notice if parts other than original Kohler Co. parts or spare parts purchased from Kohler Co. are used for maintenance and repair.

Under extreme conditions, more frequent maintenance than scheduled in the maintenance and inspection schedule can be required.

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For further information, contact your nearest authorized Kohler customer service.

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List of changes

Version	From serial number	Comment	Date
33521029401_1_1	2016 16 0001	First version	December 2016
33521029401_2_1	2016 16 0001	Various amendments	April 2017
33521029401_3_1	2016 16 0001	Change of maintenance schedule	August 2017
33521029401_4_1	2016 16 0001	Various amendments	August 2018
33521029401_5_1	2016 16 0001	Corrections	November 2018
33521029401_6_1	2016 16 0001	Maintenance schedule updated (exhaust aftertreatment system maintenance activities integrated). Service chapter removed. Warranty statement for emission control integrated.	March 2021
33521029501_7_1	2016 16 0001	Change of the graphics in the chapter “Design overview”. Change of the graphics in the chapter “Overview of sensors”. Addition to “Low-load operation” information in the chapter “Starting the engine”. Update of “Operating fluids and maintenance” chapter. Integration of EATS in the approved fuel qualities. Addition to description “Paraffinic fuels (according to EN15940). Addition to HVO table. Correction to “Minimum requirement for quality” table. Structural adjustment of “Maintenance” chapter. Addition to “Definition of service levels” chapter. Update of maintenance schedule. Update of “Tightening torques” chapter.	September 2022

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1 Product description

1.1 Technical description

1.1.1 Engine components

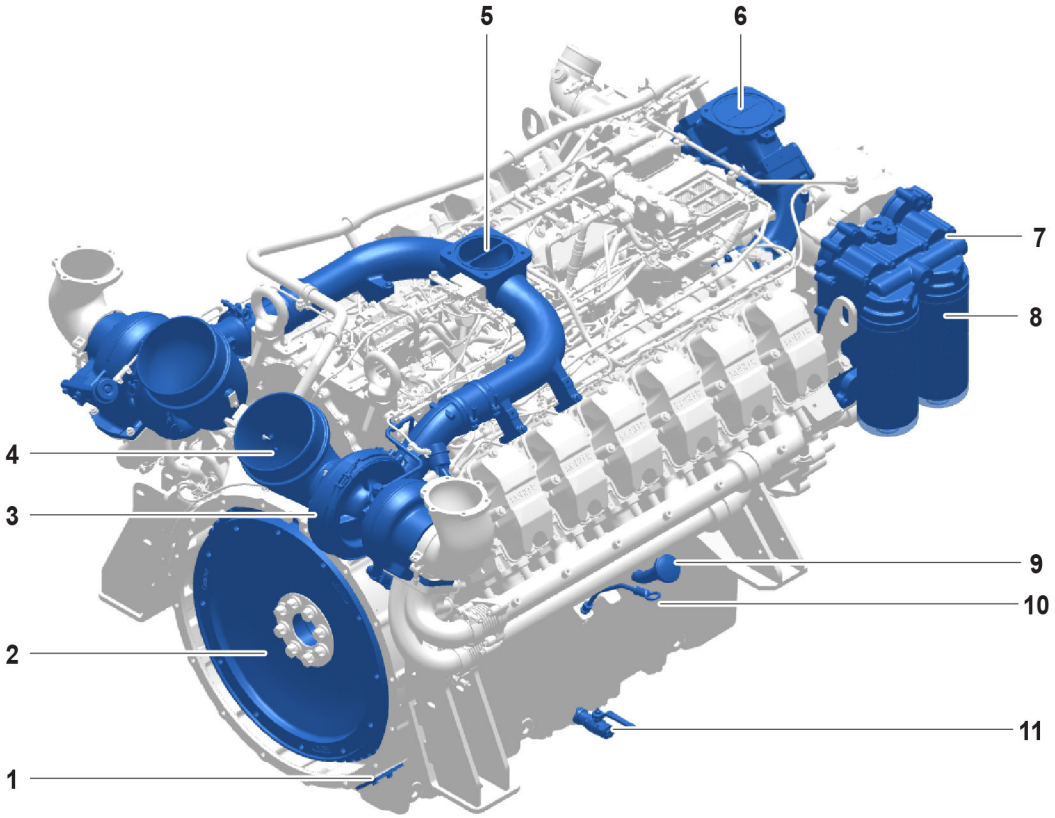

Fig. 1: Driving end engine components

1	Fastening point for turning device	**7**	Oil module
2	Flywheel	**8**	Oil filter
3	Exhaust gas turbocharger	**9**	Oil filler neck
4	Intake pipe	**10**	Oil dipstick
5	Air outlet to the charge air cooler	**11**	Oil drain stop valve
6	Air inlet from charge air cooler		

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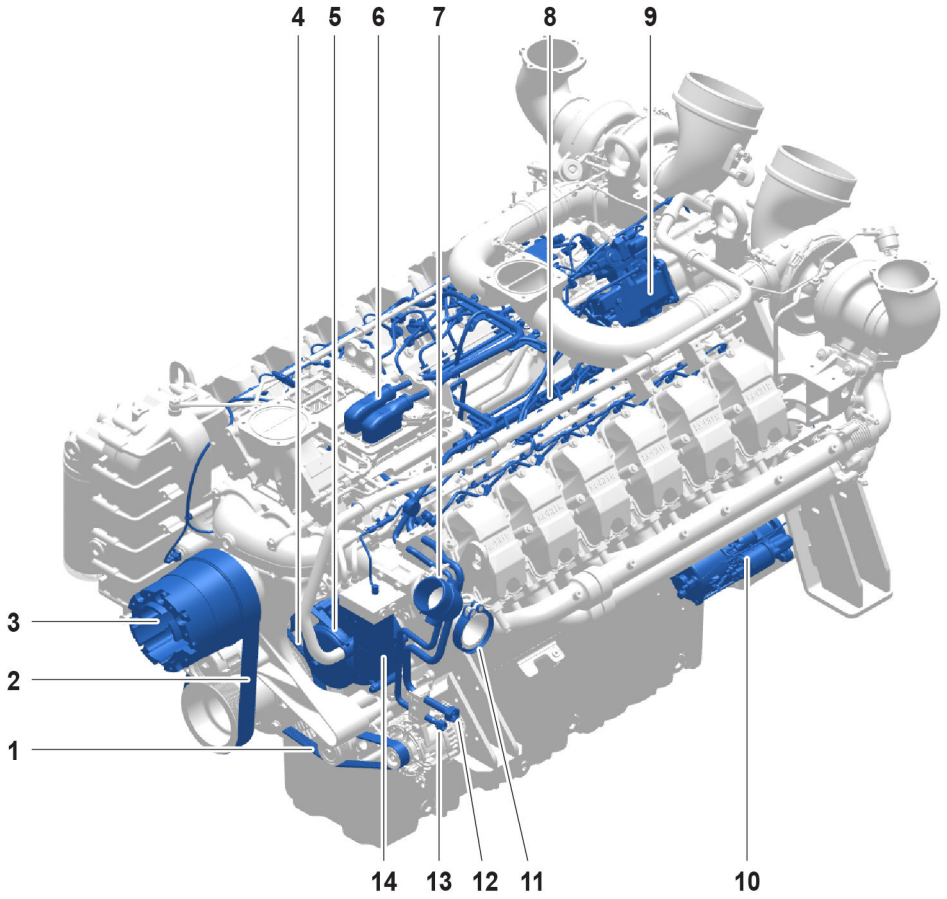

Fig. 2: Free end engine components

1	V-ribbed belt for battery charging alternator and coolant pump	**8**	Fuel rail (2x)
2	Fan drive V-ribbed belt	**9**	Fuel high pressure pump
3	Fan drive	**10**	Starter motor
4	Crankcase breather system V-ribbed belt	**11**	Coolant inlet
5	Crankcase breather system	**12**	Fuel supply
6	Engine control unit	**13**	Fuel return
7	Coolant outlet	**14**	Main fuel filter

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1.1.2 Overview of sensors

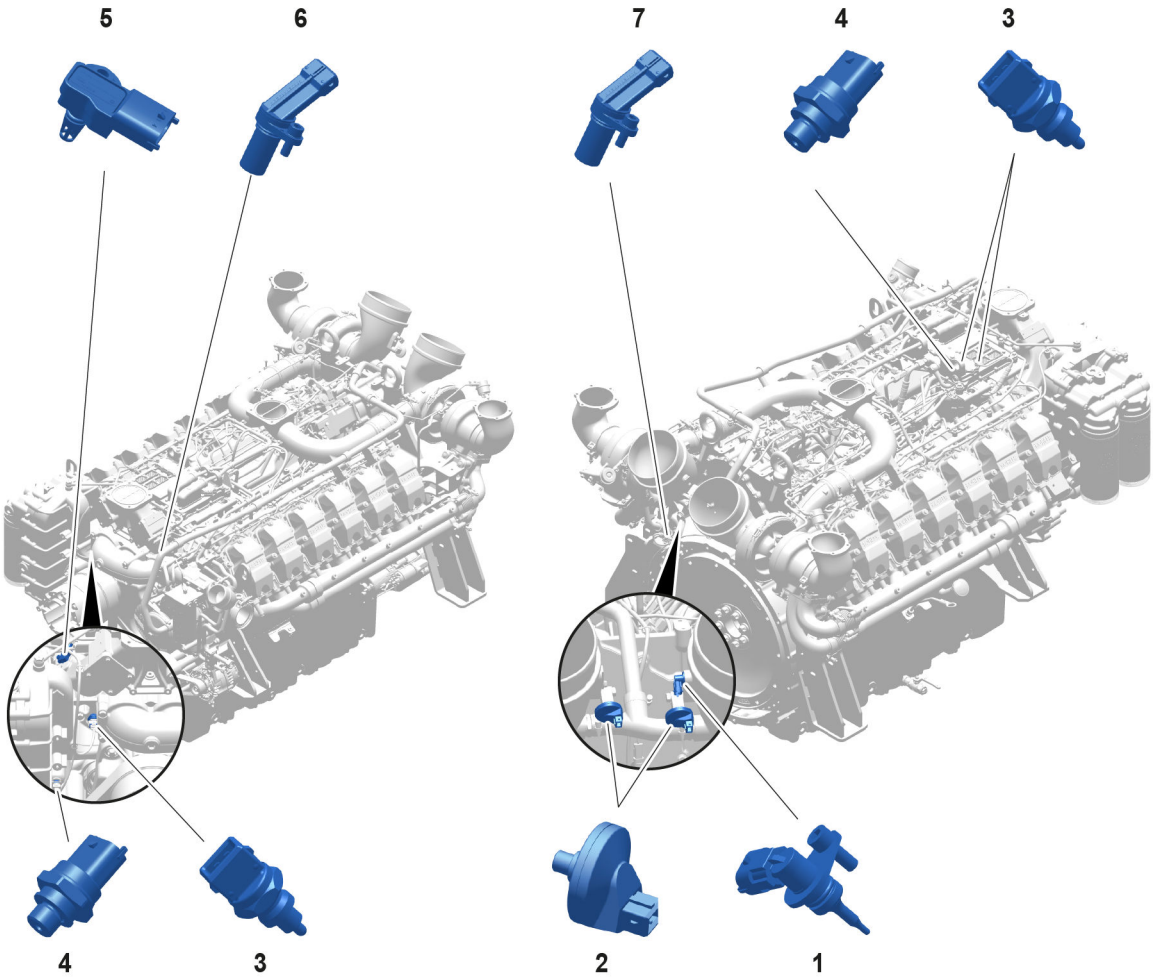

Fig. 3: Overview of sensors

1 Temperature sensor
2 Charge air pressure & temperature sensor maintenance switch (2x)
3 Temperature sensor
4 Pressure sensor
5 Pressure & temperature sensor
6 Speed sensor
7 Speed sensor

1.1.3 Engine type itemization

Engine type designation									
K	**D**	**27**	**V**	**12**	**5**	**A**	**F**	**C**	**Description**
K									Manufacturer
	D								Engine
		27							Total displacement: 27 l
			V						V-engine
				12					Number of cylinders
					5				Frequency: 5 = 50 Hz; 6 = 60 Hz

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